SMART NEST

Developer & Team Infobook

APPLICATION OVERVIEW

Turtle Tracker empowers sea turtle conservation teams to safely and efficiently relocate and monitor vulnerable nests by creating a smart artificial nest system that minimizes egg handling, preserves natural incubation conditions, and provides real-time environmental data for higher hatching success.

Core capability	What it means
One-touch relocation	Transfer eggs into the artificial nest with minimal handling.
Smart nest replication	Replicate key natural nest conditions in a controlled artificial nest.
Remote monitoring	Track temperature and environmental conditions without disturbing the nest.
Better patrol workflow	Support night patrol and collection teams with a clear digital workflow.
Data-driven conservation	Turn nest and hatching data into actionable conservation insights.

Product flow: Natural nest → One-time egg transfer → Smart artificial nest → Remote monitoring → Hatching record

PRODUCT SCOPE

Turtle Tracker connects physical nest relocation with digital monitoring. The app records where each nest is located, which eggs were collected, how they are incubated, when hatching is expected, and what actually happened at the end of incubation. This infobook defines the shared data structure and calculation rules for the whole team.

TURTLE TRACKER

Developer & Team Infobook

Hatchery setup • Nest registration • Egg collection • Incubation dates • Hatching results

THE GOAL

Keep every nest and egg batch traceable from collection to hatching, while making all date calculations consistent and easy to understand for both developers and field teams.

How to use this infobook

- Use Sections 1–3 when configuring a hatchery and its nest layout.
- Use Section 4 when registering a new egg collection.
- Use Section 5 when the hatch date is reached and the outcome is recorded.
- Use Section 6 as the implementation reference for automatic calculations.

1. Hatchery & Section Setup

The hatchery setup defines the available area, the section geometry and the spacing used to place nests. The app should use these values consistently when calculating how many nests can fit in each section.

Required information

Parameter	Current project value / input	How it is used
Incubator / hatchery area	User enters usable area in m^2	Defines the available incubation space.
Preferred spacing	70 cm center-to-center	Defines the planning grid between nests.
Nest planning area	0.49 m^2 per nest	70 cm × 70 cm planning area.
Section area	2 m × 2 m = 4 m^2	Defines one standard section.
Distance from wall	35 cm	Keeps nests away from the section boundary / wall.
Number of nests per section	Calculated	Displayed automatically from the section layout.

Section registration form

Field	Input
Shape	User selects / defines the section shape.
Dimensions	User enters the dimensions.
Start point	User defines the reference / starting point for the layout.
Incubator area	Calculated or stored from the configured geometry.
Preferred spacing	70 cm center-to-center.
Distance from wall	35 cm.

SPACING DEFINITION

70 cm means center-to-center distance between neighboring nests. The 0.49 m^2 value is a planning approximation (0.70 m × 0.70 m), not the biological diameter or physical volume of a turtle nest.

2. Temperature Reference

Temperature is a core incubation parameter. The values below are the current project guidance supplied by the team.

Temperature & sex determination

Temperature	Expected tendency	Use in the app
≈26–28 °C	Predominantly males	Lower-temperature reference
Around 29 °C	Approximately 1:1 male-to-female	Pivotal-temperature reference
≈30–32 °C	Predominantly females	Higher-temperature reference

Incubation temperature

Setting	Value
Recommended	29–30 °C
Acceptable range	26–32 °C
High-temperature warning	Above 32–33 °C may reduce hatching success and increase embryo mortality.

BIOLOGICAL NOTE

The uploaded conservation guide confirms that incubation temperature is important for the balance between male and female hatchlings. Exact sex-determination thresholds can vary by species and population, so these project values should remain configurable rather than being hard-coded as universal biological constants.

3. Incubation Period & Date Logic

The uploaded conservation guide states that incubation time depends on temperature: higher temperatures shorten the incubation period. It gives the following approximate reference values.

Average temperature	Approximate incubation period
26 °C	80–82 days
30 °C	≈56 days
32 °C	45–48 days

The date fields

Date / value	Definition	Who / what sets it
Collection Date	Date eggs are collected from the natural nest.	User
Incubation Start Date	Date incubation in the artificial nest actually begins.	System / user
Estimated Hatch Date	Predicted hatch date based on the incubation estimate.	System
Actual Hatching Date	Exact date hatching is observed / recorded.	User
Incubation Duration	Actual number of calendar days from incubation start to actual hatching.	System

DATE RULE — IMPORTANT

For accurate date calculations, the system needs a clear incubation start date. Ideally, this is the date the eggs are placed into the artificial nest. If the current prototype uses Collection Date as the incubation start, document that convention consistently throughout the app.

Estimated hatch date

The system should calculate the estimated hatch date by adding the configured expected incubation duration to the incubation start date.

Estimated Hatch Date = Incubation Start Date + Estimated Incubation Duration

Example: at a 30 °C reference duration of approximately 56 days, an incubation start date is followed by an estimated hatch date 56 calendar days later.

Actual incubation duration

Actual Incubation Duration = Actual Hatching Date – Incubation Start Date

Keep the estimated hatch date and actual hatching date as separate values. Never overwrite the original estimate.

4. Registration Form — Before / Upon Gathering Eggs

This form creates the traceable record for one collected egg batch. The bucket, nest and section should remain linked throughout the incubation period.

Field	Input / system behavior
Bucket ID	QR generated automatically OR entered manually.
Nest Number	Assigned automatically by the system.
Section	Selected from a picker.
Number of Eggs	User enters the number of collected eggs.
Collection Date	User enters the collection date.
Check Date	User enters the relevant check date.
Estimated Hatch Date	Calculated automatically.
Location	Captured / selected using the map.

Founder	User enters the responsible person / source information.

User-facing registration form

Field	Entry
Bucket ID	☐ QR automatic ☐ Manual: ____________________________
Nest Number	Automatic: ___________________
Section	Select from picker: ___________________
Number of Eggs	___________________
Collection Date	____ / ____ / ______
Check Date	____ / ____ / ______
Estimated Hatch Date	Automatic: ____ / ____ / ______
Location	Map / coordinates
Founder	______________________________

5. Hatching Day — Final Results

When hatching is completed or the nest is checked after the expected hatch window, the team records the actual outcome.

Field	Input / calculation
Actual Hatching Date	User records the exact observed date.
Incubation Duration	Automatic: Actual Hatching Date − Incubation Start Date.
Rotten Eggs Number	User enters the number of rotten eggs.
Hatched Egg Number	User enters the number of eggs that hatched.
Not Hatched Eggs	Automatic.
Hatching Egg Percentage	Automatic.

Automatic calculations

1. Not Hatched Eggs

Not Hatched Eggs = Number of Eggs − Rotten Eggs − Hatched Eggs

2. Hatching Egg Percentage

Hatching Egg Percentage = Hatched Eggs ÷ Number of Eggs × 100

CURRENT METRIC DEFINITION

For the current prototype, Hatching Egg Percentage uses the total number of collected eggs as the denominator. If the team later wants a separate viability-adjusted metric that excludes rotten eggs, it should be stored as a different metric rather than changing this calculation.

6. Record Status & User Experience

The record should move through a simple lifecycle so that users and developers always know what has happened.

Stage	Key information	System status
Registered	Bucket + nest + section + egg count	Collection recorded
Incubating	Start date + temperature + estimated hatch date	Incubation active
Check due	Check date / expected hatch window	Action required
Hatched	Actual hatch date + hatched eggs	Hatching recorded
Completed	Rotten + hatched + not hatched + percentage	Record complete

7. Developer Data Structure

Recommended relationship between the records:

Hatchery → Section → Nest → Bucket / Egg Collection Record → Incubation Record → Hatching Result

Entity	Important fields	Automatic behavior
Hatchery	Incubator area	Stores total usable area.
Section	Shape, dimensions, start point, wall distance	Calculates / displays nest capacity.
Nest	Nest number, section	Nest number assigned automatically.
Bucket	Bucket ID	QR-generated or manual.
Egg Collection	Egg count, collection date, check date, location, founder	Creates incubation record.
Incubation	Start date, temperature / estimate, estimated hatch date	Calculates estimated hatch date.
Hatching Result	Actual hatch date, rotten, hatched	Calculates duration, not-hatched count and hatching percentage.

8. Developer Checklist

- Use one consistent date format across the app and store dates as real date values, not formatted text.
- Keep Collection Date, Incubation Start Date, Estimated Hatch Date and Actual Hatching Date as separate fields.
- Do not overwrite Estimated Hatch Date when Actual Hatching Date is entered.
- Calculate Incubation Duration only after an Actual Hatching Date is available.
- Prevent Not Hatched Eggs from becoming negative.
- Prevent Hatched Eggs + Rotten Eggs from exceeding Number of Eggs.
- If the user changes the incubation temperature / estimate, recalculate the Estimated Hatch Date but preserve the previous actual result.
- Nest Number should be system-generated; Bucket ID may be QR-generated or entered manually.
- Section should be selected from the existing section list rather than typed freely.
- Keep the hatching percentage calculation transparent and consistent.

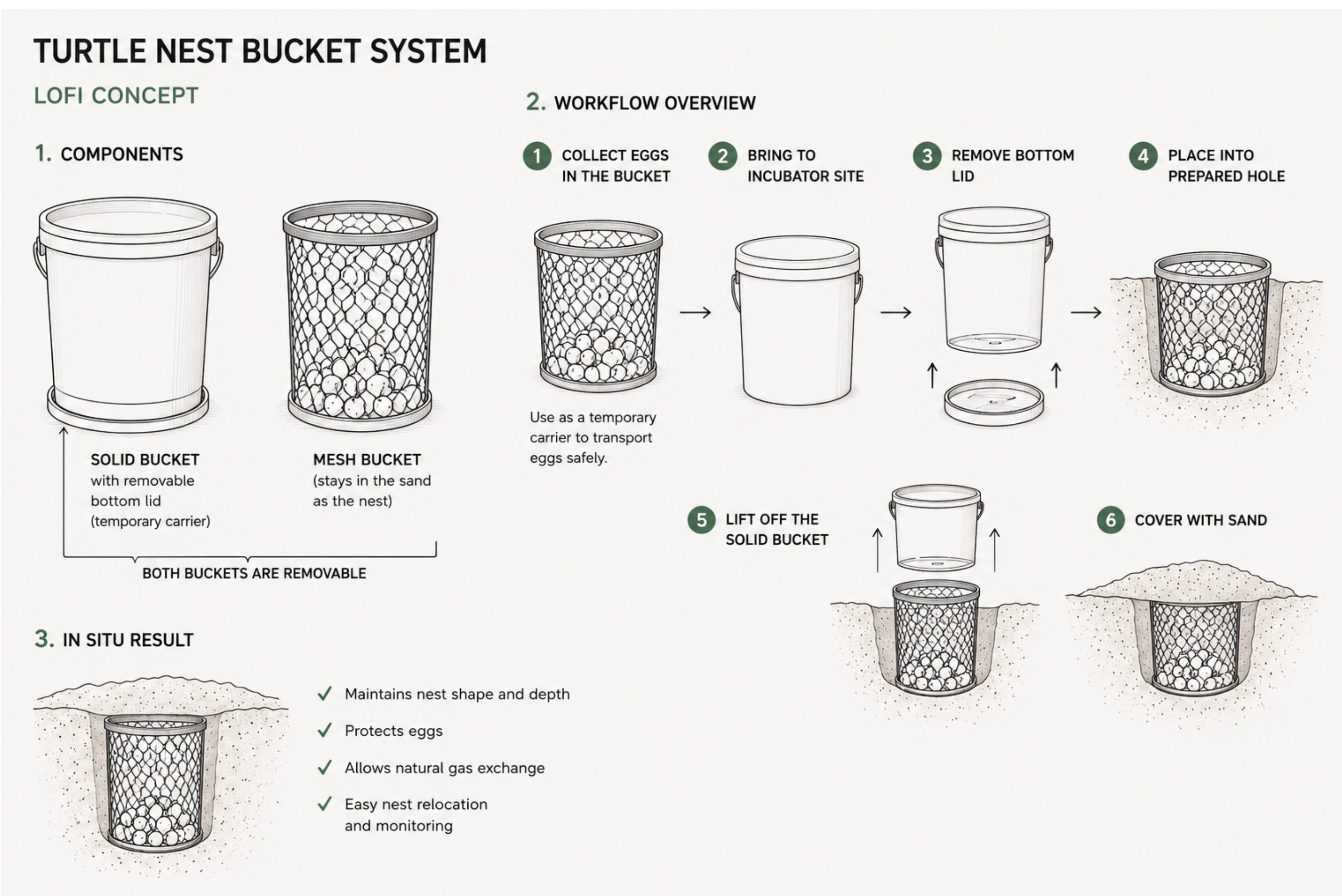
9. Physical Device & IoT

Turtle Tracker is not only a mobile application. The digital system is connected to a physical smart-nest device that stays with the artificial nest during incubation. The physical layer collects environmental data while the app provides monitoring, history and alerts without requiring unnecessary disturbance of the eggs.

9.1 Physical nest system (Figure 1)

Component	Role
Solid outer bucket	Temporary carrier and structural support during relocation.
Removable bottom lid	Allows the system to be positioned over the prepared nest opening and the solid carrier to be removed.
Mesh inner bucket	Remains around the eggs after installation and becomes part of the artificial nest.
Eggs + surrounding sand	Transferred once into the mesh nest and then covered with incubation sand.
Sensor module	Measures environmental conditions around the artificial nest.

Figure 1



9.2 IoT monitoring layer

Data	Purpose	App behavior
Sand / nest temperature	Monitor incubation conditions and temperature trends.	Display current value, history and status.
Moisture / humidity	Monitor the moisture condition of the artificial nest.	Display current value, history and warnings when configured limits are exceeded.
Timestamp	Associate every sensor reading with a point in time.	Build the incubation timeline and historical charts.
Nest ID	Connect sensor data to the correct physical nest.	Show data inside the corresponding nest record.
Section / location	Connect the device to its physical position in the hatchery.	Allow the team to identify the monitored nest on the hatchery layout / map.

9.3 Device → App data flow

Physical Nest → Sensors → IoT Device → Data Transmission → Turtle Tracker App → Dashboard / History / Alerts

ONE NEST = ONE DIGITAL RECORD

Each physical smart nest should have a unique digital identity. Sensor readings must be associated with the correct Nest Number / Bucket ID so that temperature and moisture history cannot be mixed between nests.

9.4 Hardware requirements for the prototype

- Temperature sensor suitable for continuous sand / nest monitoring.
- Moisture or humidity sensor suitable for the intended sand environment.
- Microcontroller / IoT board capable of reading the sensors.
- Power source appropriate for the full incubation period.
- Wireless communication method supported by the prototype environment.
- Protective enclosure for electronics that keeps the sensor system safe from sand and moisture while allowing the sensors to measure the nest environment.
- A unique device identifier linked to the corresponding nest record in the app.

9.5 Monitoring logic

Condition	System response
Normal range	Continue monitoring and store readings.
Temperature / moisture approaching configured limit	Show Attention status and notify the user according to the configured alert logic.
Critical threshold exceeded	Show Critical status and trigger an urgent alert if notifications are enabled.
No recent sensor data	Show No Data status rather than displaying a stale value as current.
Device disconnected	Record the last successful reading and show the communication status.

9.6 Important implementation principle

The IoT system should support conservation work rather than increase disturbance. The purpose of remote monitoring is to reduce the need to open, move or physically inspect the nest simply to obtain environmental data. The sensor system should therefore be designed for stable long-term monitoring with minimal interaction.

BIOLOGICAL CONTEXT

Environmental conditions such as sand temperature and moisture are important to incubation success. The uploaded conservation guide notes that artificial incubation should reproduce appropriate natural incubation conditions and that temperature strongly affects incubation duration and sex ratio. ?filecite?turn3file0?L1-L1?

10. Source & Scope

The incubation-period reference values in this document come from the uploaded "Conservation of Sea Turtles — 101 Questions and Answers". The source states that eggs hatch within approximately 26–32 °C and gives approximately 80–82 days at 26 °C, 56 days at 30 °C and 45–48 days at 32 °C.

The same source states that artificial incubation should be performed by trained personnel and notes that moving eggs to an artificial incubation area within approximately 3 hours after laying does not significantly affect hatching ability in the cited studies; after that, eggs require more careful handling.

Hatchery layout values (70 cm spacing, 0.49 m^2 planning area, 2 m × 2 m sections and 35 cm wall distance) are Turtle Tracker project specifications supplied by the team.

This infobook is an implementation and team-reference document. Local conservation authority requirements and species-specific protocols take precedence where applicable.